PHYS 365 — Final Exam Fall 2024 — Worked Solutions

30 questions. Every step uses the formula sheet printed at the back of the F2024 final. Anything that is *not* on that sheet is put in an orange **REMEMBER** box. Every final answer was checked against the official answer key. Note: this sheet uses $\Delta x\,\Delta p_x \geq \dfrac{h}{2\pi}$. The F2025 sheet uses $\dfrac{h}{4\pi}$, so use whichever version your exam sheet gives.

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	6	C	11	A	16	D	21	C	26	C
2	A	7	B	12	A	17	C	22	A	27	C
3	C	8	D	13	B	18	B	23	D	28	B
4	B	9	B	14	A	19	A	24	C	29	C
5	A	10	D	15	C	20	C	25	B*	30	A

NOTE ON THE ANSWER KEY

*Q25: the stopping voltage is misprinted as 4.57×10^{19} V, which is physically impossible. The value that was intended is $K_{max} \approx 7.32$ eV. See the full solution.

REMEMBER — not on the formula sheet

- Pressure amplitude from intensity: $I = \dfrac{(\Delta p_m)^2}{2\rho v}$. Combine $I = \frac{1}{2}\rho v \omega^2 s_m^2$ with $\Delta p_m = v\rho\omega s_m$ (Q1).
- Power transmission written with $r = \dfrac{Z_2}{Z_1}$: $T_e = \dfrac{4r}{(1+r)^2}$ (Q2).
- Standing wave with n loops: $n+1$ nodes and n antinodes (Q3).
- Open end = pressure node (displacement antinode). Closed end = displacement node (pressure antinode) (Q4).
- Double-slit fringe spacing (small angles): $\Delta y = \dfrac{\lambda D}{d}$ (Q6).
- Single slit: the central maximum is twice as wide as each side maximum (Q7).
- A function $f(x+vt)$ moves in $-x$ at speed v. A function $f(x-vt)$ moves in $+x$ (Q9).
- SHM: $a = \dfrac{d^2x}{dt^2}$. For $x = A\sin(\omega t + \phi)$, $a = -A\omega^2 \sin(\omega t + \phi)$ (Q10).
- Noise cancelling: the emitted wave must be π out of phase with the incoming wave at the moment it is emitted (Q13).
- $K = E - U,\ p = mv,\ K = \dfrac{p^2}{2m}$ (Q19, Q20, Q26, Q29).
- Photon energy $E = \dfrac{hc}{\lambda}$. Photon momentum $p = \dfrac{h}{\lambda} = \dfrac{hf}{c}$. Momentum is conserved in absorption (Q21, Q26, Q29).
- "k-th excited state" means $n = k+1$ (Q21, Q23, Q24).
- Probability density $\propto$ intensity (Q30).

Question 1 Pressure amplitude of a whisper

1. The sound level made by a whispering person is 30 dB. What is the pressure amplitude of the sound assuming a frequency of 1000 Hz, a density of air of 1.2 $\mathrm{kg/m^3}$, and a speed of sound of 343 $\mathrm{m/s}$.

   A) 9.1×10^{-4} Pa
   B) 1.3×10^{-4} Pa
   C) 2.0×10^{-9} Pa
   D) 3.5×10^{-10} Pa

FROM THE FORMULA SHEET

$$\beta = (10\ \mathrm{dB}) \log_{10}\left(\frac{I}{I_0}\right), \quad I = \frac{1}{2}\rho v \omega^2 s_m^2, \quad \Delta p_m = (v\rho\omega) s_m$$

REMEMBER — not on the formula sheet

Substitute $s_m = \dfrac{\Delta p_m}{v\rho\omega}$ into I to get $I = \dfrac{(\Delta p_m)^2}{2\rho v}$.

Step 1. Intensity: $I = I_0 10^{\beta/10} = 10^{-12} \times 10^3 = 1.0 \times 10^{-9}\ \frac{\mathrm{W}}{\mathrm{m^2}}$

Step 2. Pressure amplitude:

$$\Delta p_m = \sqrt{2\rho v I} = \sqrt{2(1.2)(343)(1.0 \times 10^{-9})} = \sqrt{8.23 \times 10^{-7}} = 9.1 \times 10^{-4}\ \mathrm{Pa}$$

Final answer: **(A)** 9.1×10^{-4} **Pa** ✓ Matches answer key

Question 2 Impedance from 81% power transmission

2. An acoustic wave of amplitude A is travelling in water, whose impedance is $Z_w = 1.48$ MPa. The wave is incident on a boundary with another liquid with impedance Z_L such that 81% of the incident wave's **power** is transmitted. What is the impedance of the other liquid?

   A) 3.77 MPa
   B) 2.17 MPa
   C) 3.84 MPa
   D) 5.81 MPa

FROM THE FORMULA SHEET

$$T = \frac{2Z_1}{Z_1 + Z_2}, \quad T_e = \frac{Z_2}{Z_1} T^2, \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Step 1. Let $r = \dfrac{Z_L}{Z_w}$. Then

$$T_e = r\left(\frac{2}{1+r}\right)^2 = \frac{4r}{(1+r)^2} = 0.81$$

Step 2. Rearrange into a quadratic: $0.81r^2 + 1.62r + 0.81 = 4r \;\Rightarrow\; 0.81r^2 - 2.38r + 0.81 = 0$

$$r = \frac{2.38 \pm \sqrt{2.38^2 - 4(0.81)^2}}{2(0.81)} = \frac{2.38 \pm 1.744}{1.62} = 2.545 \text{ or } 0.393$$

Step 3. $Z_L = 2.545(1.48) = 3.77$ MPa (the other root, 0.58 MPa, is not an option).

Final answer: **(A) 3.77 MPa** ✓ Matches answer key

Question 3 Nodes and antinodes, 5th harmonic

3. If you set up the resonance for the fifth harmonic on a string, how many nodes and antinodes are in the pattern?

   A) 5 nodes and 6 antinodes
   B) 4 nodes and 5 antinodes
   C) 6 nodes and 5 antinodes
   D) 5 nodes and 4 antinodes

FROM THE FORMULA SHEET
$f_n = \dfrac{nv}{2L}$, i.e. n half-wavelengths on the string

Step 1. The 5th harmonic has 5 loops. There is a node at each end and one between each pair of loops, so there are $5 + 1 = 6$ nodes. There is one antinode per loop, so 5 antinodes.

Final answer: **(C) 6 nodes, 5 antinodes** ✓ Matches answer key

Question 4 Pipe ends

4. An organ pipe with one end open and the other closed is operating at one of its resonant frequencies. The open and closed ends are, respectively:

   A) pressure node, pressure node
   B) pressure node, displacement node
   C) displacement antinode, pressure node
   D) displacement node, displacement node

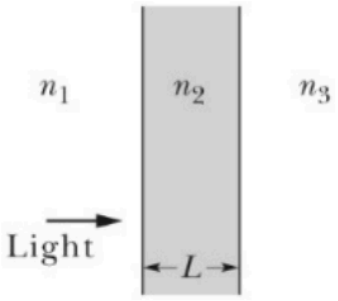
Step 1. At an **open** end the pressure is fixed at atmospheric, so it is a pressure node (and a displacement antinode).

Step 2. At a **closed** end the air cannot move, so it is a displacement node (and a pressure antinode).

Final answer: **(B) Pressure node, displacement node** ✓ Matches answer key

Question 5 Thin film, destructive reflection

5. The figure below shows a thin film with different materials on either side, where $n_1 = 1.30$, $n_2 = 1.50$, and $n_3 = 1.40$. A ray of light with a wavelength in vacuum of 600 nm is incident normal to the film. What is the second nonzero smallest value of the film thickness L that produces **destructive** interference in the reflected waves?



A) 400 nm
B) 200 nm
C) 100 nm
D) 300 nm

FROM THE FORMULA SHEET

$2t = m\frac{\lambda_0}{n}$, $\quad 2t = \left(m - \frac{1}{2}\right)\frac{\lambda_0}{n}$. Reflection off a lower index gives 0. Off a higher index gives 0.5 wavelength.

Step 1. Top surface ($1.30 \rightarrow 1.50$, off a higher index): $\frac{1}{2}\lambda$. Bottom surface ($1.50 \rightarrow 1.40$, off a lower index): 0. The net shift is half a wavelength.

Step 2. With a net half-wave shift, a path difference of a whole number of wavelengths gives **destructive** interference:

$$2L = m\frac{\lambda_0}{n_2} \;\Rightarrow\; L = \frac{m\lambda_0}{2n_2} = \frac{m(600)}{2(1.50)} = 200m \text{ nm}$$

Step 3. Nonzero values: 200 nm ($m = 1$) and 400 nm ($m = 2$). The second smallest is 400 nm.

Final answer: **(A) 400 nm** ✓ Matches answer key

Question 6 Keeping the fringe spacing

6. In a Young's double-slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance D must be changed to

A) $D/2$
B) $D\sqrt{2}$
C) $2D$
D) $4D$

FROM THE FORMULA SHEET
$d \sin\theta = m\lambda$

REMEMBER — not on the formula sheet
Small angles: $y_m = \dfrac{m\lambda D}{d}$, so the spacing is $\Delta y = \dfrac{\lambda D}{d}$.

Step 1. $\Delta y = \dfrac{\lambda D}{d}$. Doubling d halves Δy, unless D is also doubled:

$$\frac{\lambda(2D)}{2d} = \frac{\lambda D}{d}$$

Final answer: **(C)** $2D$ ✓ **Matches answer key**

Question 7 Single-slit pattern shape

7. Monochromatic plane waves of light are incident normally on a single slit. Which one of the four figures below most correctly shows the diffraction pattern observed on a distant screen?

A) 

B) 

C) 

D) 

FROM THE FORMULA SHEET
$a \sin\theta = m\lambda$, $m = 1, 2, 3, \ldots$ (minima)

Step 1. The minima are equally spaced (for small angles) at $m = \pm 1, \pm 2, \ldots$. The central bright band runs from $m = -1$ to $m = +1$, so it is **twice** as wide as each side band, and the side bands all have equal width.

Step 2. Only pattern B shows a central bright band twice the width of equal side bands.

Final answer: **(B) Pattern B** ✓ **Matches answer key**

Question 8 Phase of a phasor

8. A phasor $\tilde{A}$ expressed in cartesian form has the following real and imaginary components:

$$Re[\tilde{A}] = -7.25 \text{ mm}$$
$$Im[\tilde{A}] = +4.38 \text{ mm}$$

What is the phase constant of the phasor?

A) -0.543 rad
B) -1.03 rad
C) 2.11 rad
D) 2.60 rad

FROM THE FORMULA SHEET
$\tilde{A} = A(\cos\theta + j\sin\theta)$

Step 1. Re < 0 and Im > 0, so the phasor is in the 2nd quadrant:

$$\theta = \pi - \tan^{-1}\left(\frac{4.38}{7.25}\right) = \pi - 0.543 = 2.60 \text{ rad}$$

(-0.543 is the calculator's answer, which points into the 4th quadrant. That is the trap.)

Final answer: **(D) 2.60 rad** ✓ Matches answer key

Question 9 Pulse moving in $-x$ at 20 $\frac{\text{m}}{\text{s}}$

9. A wave pulse is travelling through a medium in the $-x$-direction at a speed of 20 m/s. Which of the following functions could possibly describe the propagating pulse?

A) $f(x,t) = 20(x+t)^2$
B) $f(x,t) = 5e^{-(40t+2x)^2}$
C) $f(x,t) = \frac{50}{60t-3x}$
D) $f(x,t) = 200t^2 + 10x^2$

FROM THE FORMULA SHEET
$y(x,t) = f(x \pm vt)$

Step 1. We need the form $f(x+20t)$.

Step 2. Check B: $40t + 2x = 2(x+20t)$ ✓. A has $x+t$, which is speed 1. C has $60t - 3x = -3(x-20t)$, which moves in $+x$. D is not a function of $x \pm vt$.

Final answer: **(B)** $5e^{-(40t+2x)^2}$ ✓ Matches answer key

Question 10 Acceleration of an SHM

10. The position of an object undergoing simple harmonic motion is given by:

$$x(t) = A\sin(\omega t + \pi/4)$$

Which expression below gives the acceleration of the object?

A) $a(t) = -A\omega^2(\sin(\omega t) + \cos(\omega t))$

B) $a(t) = A\omega^2\cos(\omega t + \pi/4)$

C) $a(t) = \frac{A\omega^2}{\sqrt{2}}(\sin(\omega t) + \cos(\omega t))$

D) $a(t) = A\omega^2\cos(\omega t + 3\pi/4)$

FROM THE FORMULA SHEET

$\cos\left(\alpha \pm \frac{\pi}{2}\right) = \mp\sin\alpha$

REMEMBER — not on the formula sheet

Differentiate twice: $a = \dfrac{d^2x}{dt^2} = -A\omega^2\sin\left(\omega t + \frac{\pi}{4}\right)$

Step 1. Use the identity with $\alpha = \omega t + \frac{\pi}{4}$:

$$-\sin\alpha = \cos\left(\alpha + \frac{\pi}{2}\right) \Rightarrow a = A\omega^2\cos\left(\omega t + \frac{\pi}{4} + \frac{\pi}{2}\right) = A\omega^2\cos\left(\omega t + \frac{3\pi}{4}\right)$$

Final answer: **(D)** $A\omega^2\cos\left(\omega t + \frac{3\pi}{4}\right)$ ✓ **Matches answer key**

Question 11 Pendulum on the Moon

11. A simple pendulum is made from a ball of mass m hanging from a string of length ℓ and when it is set swinging on Earth it is observed to have period T. The pendulum is then taken to the moon where the acceleration due to gravity is $1/6$ that on Earth. What is the period of the pendulum on the moon?

A) It is $\sqrt{6}$ times longer than on Earth
B) It is 6 times longer than on Earth
C) It is $1/6$ as long as on Earth
D) It is the same as on Earth

FROM THE FORMULA SHEET

$T = 2\pi\sqrt{\dfrac{\ell}{g}}$

$$\frac{T_{moon}}{T} = \sqrt{\frac{g}{\frac{g}{6}}} = \sqrt{6}$$

Final answer: **(A)** $\sqrt{6}$ **times longer** ✓ **Matches answer key**

Question 12 Phase constant from the resultant amplitude

12. The superposition of two waves of equal amplitude and opposite phase constants,

$$y_1(x,t) = A\sin(kx - \omega t + \phi_0)$$
$$y_2(x,t) = A\sin(kx - \omega t - \phi_0)$$

yields the resultant wave $y_r(x,t) = \sqrt{3}A\sin(kx - \omega t)$. What is the phase constant ϕ_0 that produces this resultant wave?

A) $\pi/6$
B) $3\pi/4$
C) $\pi/3$
D) $\pi/8$

FROM THE FORMULA SHEET

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

Step 1. With $\alpha = kx - \omega t + \phi_0$ and $\beta = kx - \omega t - \phi_0$:

$$y_1 + y_2 = 2A\cos(\phi_0)\sin(kx - \omega t)$$

Step 2. Set the amplitude equal to $\sqrt{3}A$:

$$2A\cos\phi_0 = \sqrt{3}A \;\Rightarrow\; \cos\phi_0 = \frac{\sqrt{3}}{2} \;\Rightarrow\; \phi_0 = \frac{\pi}{6}$$

Final answer: **(A)** $\frac{\pi}{6}$ ✓ **Matches answer key**

Question 13 Noise-cancelling phase

13. Noise-cancelling headphones detect a sound wave of frequency 440 Hz. It takes the headphones 2.00 ms to register the signal and emit a wave with the same amplitude and frequency. Taking the phase constant of the initial wave to be zero, what should the phase constant of the emitted wave be to cancel the noise?

A) -0.76π
B) 0.76π
C) 1.76π
D) -0.24π

FROM THE FORMULA SHEET

$$\omega = 2\pi f$$

Step 1. The headphones replay the wave delayed by Δt, so the emitted wave is $A\cos(\omega(t-\Delta t)+\phi)$. To cancel the noise it must equal $-A\cos(\omega t) = A\cos(\omega t+\pi)$. So

$$-\omega\Delta t + \phi = \pi \;\Rightarrow\; \phi = \pi + \omega\Delta t$$

Step 2. Phase delay: $\omega\Delta t = 2\pi(440)(2.00\times10^{-3}) = 1.76\pi$

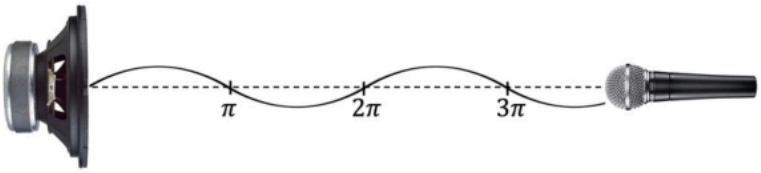
$$\phi = \pi + 1.76\pi = 2.76\pi \equiv 2.76\pi - 2\pi = 0.76\pi$$

Final answer: **(B)** 0.76π ✓ Matches answer key

Question 14 Air temperature from phase difference

14. A speaker and a microphone are placed 1.00 m apart. The frequency of the emitted wave is 600 Hz and the phase difference between the emitted and detected wave is $\Delta\phi = 3.75\pi$, as shown in the diagram. What is the temperature of the air?
(Hint: air is made of a mixture of N_2 and O_2 with a molecular mass of 28.8 g/mol)

A) $-20°C$
B) $+20°C$
C) 212 K
D) 273 K



FROM THE FORMULA SHEET

$$\Delta\phi = \frac{\Delta L}{\lambda}2\pi, \quad v = f\lambda, \quad v = \sqrt{\frac{\gamma RT}{M}}, \quad \gamma_{diatomic} = \frac{7}{5}, \quad T_K = T_{°C} + 273.15$$

Step 1. Wavelength:

$$3.75\pi = \frac{1.00}{\lambda}2\pi \;\Rightarrow\; \lambda = \frac{2}{3.75} = 0.533\ \text{m}$$

Step 2. Speed: $v = f\lambda = 600(0.533) = 320\ \frac{\text{m}}{\text{s}}$

Step 3. Temperature:

$$T = \frac{v^2 M}{\gamma R} = \frac{(320)^2(0.0288)}{1.4(8.314)} = 253.4\ \text{K} = -19.8°\text{C} \approx -20°\text{C}$$

Final answer: **(A)** $-20°$**C** ✓ Matches answer key

Question 15 Acoustic impedance vs. altitude

15. Within the first few kilometers above sea level, the atmospheric pressure p and air density ρ vary with altitude h according to

$$p(h) = p_0\left(1 - \frac{h}{\alpha}\right) \qquad \text{and} \qquad \rho(h) = \rho_0\left(1 - \frac{h}{\beta}\right)$$

where $\alpha = 8.4$ km, $\beta = 10.4$ km, and p_0 and ρ_0 are the air pressure and density at sea level. How does the acoustic impedance of air change with altitude near Earth's surface?

Hint: $\Delta\rho = \Delta\left(\frac{M}{V}\right) = \frac{\partial}{\partial V}\left(\frac{M}{V}\right)\Delta V = \left(-\frac{M}{V^2}\right)\Delta V = -\frac{\rho}{V}\Delta V$, so you can use $\frac{\Delta V}{V} = -\frac{\Delta\rho}{\rho}$.

A) It is approximately constant: $Z(h) = Z_0$

B) It decreases slowly with h: $Z(h) = Z_0\sqrt{1 - \frac{h}{\beta}}$

C) It decreases linearly with h: $Z(h) = Z_0\left(1 - \frac{h}{\beta}\right)$

D) It decreases quadratically with h: $Z(h) = Z_0\left(1 - \frac{h}{\beta}\right)^2$

FROM THE FORMULA SHEET

$Z = \sqrt{\rho B}$, $\quad B = -\dfrac{\Delta p}{\left(\frac{\Delta V}{V}\right)}$, with the hint $\dfrac{\Delta V}{V} = -\dfrac{\Delta\rho}{\rho}$

Step 1. Using the hint, $B = \rho\dfrac{\Delta p}{\Delta\rho}$. From the given profiles: $\Delta p = -\dfrac{p_0}{\alpha}\Delta h$ and $\Delta\rho = -\dfrac{\rho_0}{\beta}\Delta h$. Divide:

$$\frac{\Delta p}{\Delta\rho} = \frac{p_0\beta}{\rho_0\alpha} = \text{constant} \;\Rightarrow\; B = \rho\,\frac{p_0\beta}{\rho_0\alpha}$$

Step 2. Impedance:

$$Z = \sqrt{\rho B} = \rho\sqrt{\frac{p_0\beta}{\rho_0\alpha}} \;\propto\; \rho = \rho_0\left(1 - \frac{h}{\beta}\right) \;\Rightarrow\; Z = Z_0\left(1 - \frac{h}{\beta}\right)$$

Final answer: **(C) Decreases linearly:** $Z_0\left(1 - \frac{h}{\beta}\right)$ ✓ **Matches answer key**

Question 16 Slit width

16. A laser beam with wavelength 720 nm is shone on a single slit aperture. On the detection screen 2.00 m away, the first dark fringe of the diffraction pattern is located 1.83 cm away from the central maximum. What is the width of the slit?

A) 118 μm
B) 39.3 μm
C) 59.0 μm
D) 78.7 μm

FROM THE FORMULA SHEET
$a \sin\theta = m\lambda$

REMEMBER — not on the formula sheet
Small angles: $\sin\theta \approx \dfrac{y}{D}$

$$a = \frac{\lambda D}{y_1} = \frac{(720 \times 10^{-9})(2.00)}{0.0183} = 7.87 \times 10^{-5}\ \text{m} = 78.7\,\mu\text{m}$$

Final answer: **(D) 78.7 µm** ✓ Matches answer key

Question 17 Hubble resolution

17. The Hubble Space Telescope is imaging distant galaxies at a wavelength of 550 nm. The diameter of the telescope's objective is 2.40 m. What is the smallest angular resolution of the Hubble Space Telescope at this wavelength?

A) 1.602×10^{-5} rad
B) 3.204×10^{-5} rad
C) 2.796×10^{-7} rad
D) 5.592×10^{-7} rad

FROM THE FORMULA SHEET
$\theta_R = \sin^{-1}\left(\dfrac{1.22\lambda}{d}\right)$

$$\theta_R = \sin^{-1}\left(\frac{1.22(550 \times 10^{-9})}{2.40}\right) = 2.796 \times 10^{-7}\ \text{rad}$$

Final answer: **(C)** 2.796×10^{-7} **rad** ✓ Matches answer key

Question 18 Redshifted Lyman lines

18. The light corresponding to the Lyman spectral series of hydrogen is observed from a distant star and it appears to have a slightly longer wavelength than normal. Which statement below must be true?

A) Earth is moving toward the star.
B) The star is moving away from Earth.
C) The star is moving toward Earth.
D) The distance between Earth and the star remains the same.

Step 1. A longer wavelength means a lower frequency, so the source and observer are separating. Of the choices, only "the star is moving away from Earth" describes a receding source.

Final answer: **(B) Star moving away** ✓ Matches answer key

Question 19 Entering higher potential energy

19. A free electron in motion along the x-axis has a localized wave function. If it enters a region of space where its potential energy increases:

A) its wavelength will increase.
B) its kinetic energy will increase.
C) its wave number will increase.
D) its total energy will decrease.

FROM THE FORMULA SHEET

$$p = \frac{h}{\lambda}, \quad K = \frac{1}{2}mv^2$$

REMEMBER — not on the formula sheet

Total energy is conserved: $K = E - U$. Also $p = \sqrt{2mK}$.

Step 1. U goes up and E stays fixed, so K goes down, so p goes down, so $\lambda = \frac{h}{p}$ **increases** (and $k = \frac{2\pi}{\lambda}$ decreases).

Final answer: **(A) Wavelength increases** ✓ Matches answer key

Question 20 Speed uncertainty vs. mass

20. Three particles with masses m_1, $m_2 = 2m_1$, and $m_3 = 3m_1$ are confined in the same one-dimensional space. Measurements of the speeds of the particles are made while keeping the three particles confined in the space. Which particle can have its speed measured with the least amount of uncertainty?

A) Particle 1 with mass m_1
B) Particle 2 with mass m_2
C) Particle 3 with mass m_3
D) All particles have the same minimum uncertainty

FROM THE FORMULA SHEET

$$\Delta x \, \Delta p_x \geq \frac{h}{2\pi}$$

REMEMBER — not on the formula sheet

$\Delta p = m\Delta v$

Step 1. Same confinement (same Δx), so the minimum Δp is the same for all three. Then $\Delta v = \dfrac{\Delta p}{m}$ is smallest for the largest mass, m_3.

Final answer: **(C) Particle 3** ✓ Matches answer key

Question 21 Emission lines from $n = 3$ in a well

Questions 21 - 23 are related to the following scenario:

An electron is confined to a one-dimensional infinite potential well of width $L = 0.100$ nm.

21. If the electron is in the second excited state, what wavelengths of photons could this system emit?

A) 4.1 nm only
B) 4.1 nm, and 6.6 nm only
C) 4.1 nm, 6.6 nm, and 11 nm only
D) 11 nm only

FROM THE FORMULA SHEET

$$E_n = \frac{h^2}{8m_eL^2}n^2, \quad E = hf, \quad c = f\lambda$$

Step 1. Ground-state energy:

$$E_1 = \frac{(6.63 \times 10^{-34})^2}{8(9.109 \times 10^{-31})(1.00 \times 10^{-10})^2} = 6.03 \times 10^{-18}\ \text{J}\ (37.7\ \text{eV})$$

Step 2. From $n = 3$ the possible drops are $3 \to 1$ ($8E_1$), $3 \to 2$ ($5E_1$), and then $2 \to 1$ ($3E_1$). Use $\lambda = \dfrac{hc}{\Delta E}$ with $hc = 1.989 \times 10^{-25}$ J·m:

$$\lambda_{31} = \frac{1.989 \times 10^{-25}}{8(6.03 \times 10^{-18})} = 4.1\ \text{nm}, \quad \lambda_{32} = \frac{1.989 \times 10^{-25}}{5(6.03 \times 10^{-18})} = 6.6\ \text{nm}, \quad \lambda_{21} = \frac{1.989 \times 10^{-25}}{3(6.03 \times 10^{-18})} = 11\ \text{nm}$$

Final answer: **(C) 4.1 nm, 6.6 nm, and 11 nm** ✓ Matches answer key

Question 22 Probability between $\frac{L}{8}$ and $\frac{L}{4}$

22. If the electron is in its ground state, what is the probability of finding it between $x = L/8$ and $x = L/4$?

A) 7.84%
B) 14.1%
C) 3.92%
D) 25.0%

FROM THE FORMULA SHEET

$$P = \int_{x_1}^{x_2} |\psi|^2 dx, \quad \psi_1 = \sqrt{\tfrac{2}{L}} \sin\left(\tfrac{\pi x}{L}\right), \quad \int \sin^2(ax)\, dx = \tfrac{x}{2} - \tfrac{\sin(2ax)}{4a}$$

Step 1. Evaluate:

$$P = \frac{2}{L}\left[\frac{x}{2} - \frac{L\sin\left(\frac{2\pi x}{L}\right)}{4\pi}\right]_{L/8}^{L/4} = \left[\frac{x}{L} - \frac{\sin\left(\frac{2\pi x}{L}\right)}{2\pi}\right]_{L/8}^{L/4}$$

Step 2. Upper limit: $0.25 - \dfrac{\sin\frac{\pi}{2}}{2\pi} = 0.25 - 0.1592 = 0.0908$

Step 3. Lower limit: $0.125 - \dfrac{\sin\frac{\pi}{4}}{2\pi} = 0.125 - 0.1125 = 0.0125$

$$P = 0.0908 - 0.0125 = 0.0784 = 7.84\%$$

Final answer: **(A) 7.84%** ✓ Matches answer key

Question 23 Wave function of the third excited state

23. Which equation describes the wave function if the electron is in the third excited state?

A) $\psi(x) = \left(1.99 \times 10^{10}\ \text{m}^{-\frac{1}{2}}\right)\sin\left((1.26 \times 10^{11}\ \text{m}^{-1})x\right)$

B) $\psi(x) = \left(1.41 \times 10^{5}\ \text{m}^{-\frac{1}{2}}\right)\sin\left((9.42 \times 10^{10}\ \text{m}^{-1})x\right)$

C) $\psi(x) = \left(1.99 \times 10^{10}\ \text{m}^{-\frac{1}{2}}\right)\sin\left((9.42 \times 10^{10}\ \text{m}^{-1})x\right)$

D) $\psi(x) = \left(1.41 \times 10^{5}\ \text{m}^{-\frac{1}{2}}\right)\sin\left((1.26 \times 10^{11}\ \text{m}^{-1})x\right)$

FROM THE FORMULA SHEET

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right)$$

Step 1. The third excited state is $n = 4$, with $L = 1.00 \times 10^{-10}$ m:

$$\sqrt{\frac{2}{1.00 \times 10^{-10}}} = 1.41 \times 10^{5}\ \text{m}^{-\frac{1}{2}}, \qquad \frac{4\pi}{1.00 \times 10^{-10}} = 1.26 \times 10^{11}\ \text{m}^{-1}$$

Final answer: **(D)** $(1.41 \times 10^{5})\sin\left((1.26 \times 10^{11})x\right)$ ✓ Matches answer key

Question 24 Absorbing the longest wavelength from $n = 3$

24. A hydrogen atom is in its second excited state. To what state would it jump if it were to absorb a photon with the longest possible wavelength that it could?

A) $n = 2$
B) $n = 3$
C) $n = 4$
D) $n = 5$

FROM THE FORMULA SHEET

$$E_n = \frac{-13.6 \text{ eV}}{n^2}$$

Step 1. The longest wavelength is the smallest energy jump **upward** from $n = 3$, which is $3 \to 4$. ($n = 2$ would be emission, not absorption.)

Final answer: **(C)** $n = 4$ ✓ **Matches answer key**

Question 25 Identifying the metal (photoelectric)

25. The target of a photoelectric experiment is made from an unknown metal. When the frequency of the incident light is $f_1 = 3.00 \times 10^{15}$ Hz, the stopping voltage is measured to be $V_1 = 4.57 \times 10^{19}$ V. What metal is the target made from?

A) Silver $\Phi_{\text{Ag}} = 4.46$ eV
B) Gold $\Phi_{\text{Au}} = 5.10$ eV
C) Aluminum $\Phi_{\text{Al}} = 4.06$ eV
D) Copper $\Phi_{\text{Cu}} = 4.53$ eV

FROM THE FORMULA SHEET

$E = hf, \quad hf = K_{max} + \Phi, \quad K_{max} = eV_{stop}, \quad h = 4.136 \times 10^{-15}$ eV·s

Step 1. Photon energy: $hf = (4.136 \times 10^{-15})(3.00 \times 10^{15}) = 12.41$ eV

Step 2. Stopping energy: $eV_1 = (1.602 \times 10^{-19})(4.57 \times 10^{19}) = 7.32$ (this is the number the question intends to be in eV)

$$\Phi = 12.41 - 7.32 = 5.09 \text{ eV} \approx 5.10 \text{ eV} \Rightarrow \text{gold}$$

NOTE ON THE ANSWER KEY

A stopping voltage of 4.57×10^{19} V is impossible. It equals $\dfrac{7.32}{1.602 \times 10^{-19}}$, so the setter divided by e an extra time. The intended values are $V_1 = 7.32$ V and $K_{max} = 7.32$ eV, which give gold, matching the key.

Final answer: **(B) Gold ($\Phi = 5.10$ eV)** ✓ **Key = B (typo noted)**

Question 26 Recoil speed after absorbing a photon

26. A hydrogen atom initially at rest absorbs a photon with frequency $f = 2.923$ GHz. What is the speed of the hydrogen atom after the absorption?

A) 7.092×10^{-3} m/s
B) 4.868×10^{-9} m/s
C) 3.868×10^{-6} m/s
D) 4.424×10^{16} m/s

FROM THE FORMULA SHEET

$E = hf, \quad p = \dfrac{h}{\lambda}, \quad c = f\lambda, \quad m_p = 1.67 \times 10^{-27}$ kg

REMEMBER — not on the formula sheet

Momentum is conserved: $p_{photon} = \dfrac{hf}{c} = m_H v$. Use $m_H \approx m_p$.

$$v = \frac{hf}{m_p c} = \frac{(6.63 \times 10^{-34})(2.923 \times 10^{9})}{(1.67 \times 10^{-27})(3.00 \times 10^{8})} = 3.868 \times 10^{-6}\ \tfrac{\text{m}}{\text{s}}$$

Final answer: **(C)** $3.868 \times 10^{-6}\ \frac{\text{m}}{\text{s}}$ ✓ Matches answer key

Question 27 Largest Compton shift

27. A photon travelling in the x-direction Compton scatters off an electron. Of the scattering angles listed below, which one corresponds to the outgoing photon having the largest wavelength shift compared to the initial photon?

A) $\pi/3$
B) $3\pi/2$
C) $2\pi/3$
D) $\pi/4$

FROM THE FORMULA SHEET

$\Delta\lambda = \dfrac{h}{mc}(1 - \cos\phi)$

$$1 - \cos\phi: \quad \frac{\pi}{3}: 0.5, \quad \frac{3\pi}{2}: 1, \quad \frac{2\pi}{3}: 1.5, \quad \frac{\pi}{4}: 0.29$$

Final answer: **(C)** $\frac{2\pi}{3}$ ✓ Matches answer key

Question 28 Electron microscope resolution

28. Which of the following statements about an electron microscope is TRUE?

A) The resolving power of an electron microscope is determined exclusively by the mass of the electrons.
B) One can increase the resolving power of an electron microscope by accelerating the electrons more.
C) One can increase the resolving power of an electron microscope if instead of electrons positrons are used.
D) The resolving power of an electron microscope will decrease if instead we use protons.

FROM THE FORMULA SHEET

$$\theta_R = \sin^{-1}\left(\frac{1.22\lambda}{d}\right), \quad p = \frac{h}{\lambda}$$

Step 1. Better resolution needs a smaller $\lambda = \frac{h}{p}$. Accelerating the electrons more increases p, which decreases λ. (Positrons have the same mass, so nothing changes. Heavier protons would *improve* resolution at the same speed, so D is false.)

Final answer: **(B) Accelerate the electrons more** ✓ Matches answer key

Question 29 Matching electron and X-ray wavelengths

29. An electron beam is composed of electrons moving at a speed of $v = 8.54 \times 10^4$ m/s, which is slow enough compared to c that relativistic effects can be ignored. The diffraction picture produced by the electron beam is identical to that of a beam of x-ray photons. What is the frequency of the x-ray beam?

A) 1.24×10^{17} Hz
B) 2.27×10^{18} Hz
C) 3.52×10^{16} Hz
D) 5.01×10^{16} Hz

FROM THE FORMULA SHEET

$$p = \frac{h}{\lambda}, \quad c = f\lambda$$

REMEMBER — not on the formula sheet

$$p = mv$$

Step 1. Electron wavelength:

$$\lambda = \frac{h}{m_e v} = \frac{6.63 \times 10^{-34}}{(9.109 \times 10^{-31})(8.54 \times 10^4)} = 8.52 \times 10^{-9}\ \text{m}$$

Step 2. X-ray frequency for the same λ:

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^{8}}{8.52 \times 10^{-9}} = 3.52 \times 10^{16}\ \text{Hz}$$

Final answer: **(C)** 3.52×10^{16} **Hz** ✓ **Matches answer key**

Question 30 Probability ratio from intensity

30. A bright region of a certain diffraction picture has an intensity that is 100 times that of a darker region of the same picture: $I_d = I_b/100$. What is the ratio of the probability density of a single photon hitting the bright vs the dark regions, $\eta = P_b/P_d$?

A) $\eta = 100$
B) $\eta = 10000$
C) $\eta = 10^{-2}$
D) $\eta = 1$

Step 1. The probability of a photon landing in a region is proportional to the intensity there:

$$\eta = \frac{P_b}{P_d} = \frac{I_b}{I_d} = 100$$

Final answer: **(A)** $\eta = 100$ ✓ **Matches answer key**